ZenFlow - Enlightened Yoga Pose Classification Via Transfer Learning

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Introduction

What is Yoga Pose Classification?

- Yoga improves physical and mental well-being.
- Correct posture is important for effective yoga practice.
- Beginners often perform poses incorrectly without guidance.
- Computer Vision and Deep Learning can help detect yoga poses automatically.

Goal:

Develop a system that classifies yoga poses using deep learning.

Problem Statement

Many people practice yoga using:

- Online videos
- Self-learning
- Virtual classes

However:

❌ No real-time feedback
❌ Incorrect posture may cause injury
❌ Difficult for instructors to monitor all students

Solution:

A deep learning model that automatically detects yoga poses from images or webcam.

Project Objectives

The objectives of this project are:

- Preprocess image datasets
- Apply **Transfer Learning**
- Train a **Deep Learning model**
- Evaluate model accuracy
- Build a **Flask-based web application**
- Classify yoga poses using images and webcam video

Use Case Scenarios

🧘 Scenario 1: Yoga Practitioners

- Helps practitioners refine their poses.
- Provides feedback on **pose accuracy and alignment**.
- Enhances overall yoga practice and benefits.

👩‍🏫 Scenario 2: Yoga Instructors

- Integrate pose classification in **virtual yoga classes**.
- Provide **personalized feedback to students**.
- Improve teaching effectiveness and posture correction.

🔬 Scenario 3: Yoga Research

- Analyze yoga practice patterns.
- Study the effects of yoga on **physical and mental well-being**.
- Support **evidence-based yoga therapy research**.

Dataset Description

The dataset consists of **images of 5 yoga poses**:

- Downdog
- Goddess
- Plank
- Tree
- Warrior2

Dataset details:

- Images collected from Bing
- Cleaned and resized
- Used for training and validation

Total images: 988

📂 **Dataset Link:**
https://www.kaggle.com/datasets/ujjwalchowdhury/yoga-pose-classification

Methodology

Project workflow:

1. Data Collection
2. Image Preprocessing
3. Data Augmentation
4. Transfer Learning using Xception
5. Model Training
6. Model Evaluation
7. Flask Web Application
8. Image and Video Prediction

Model Architecture

This project uses **Transfer Learning with Xception CNN**.

Steps:

1. Load **pre-trained Xception model** (ImageNet weights).
2. Freeze convolution layers.
3. Add custom layers:
   - Flatten Layer
   - Dense Layer with Softmax
4. Train the classifier on yoga pose dataset.

Model Summary:

Parameter	Value
Total Parameters	21,363,245
Trainable Parameters	501,765
Non-trainable Parameters	20,861,480

Training Process

Training Settings:

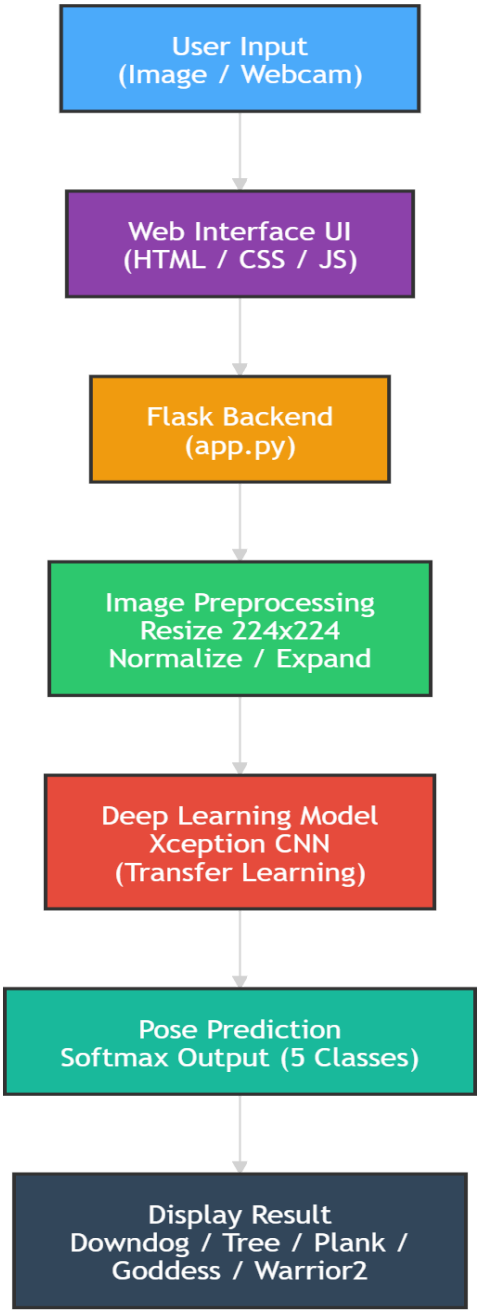
- Optimizer: Adam
- Loss Function: Categorical Crossentropy
- Epochs: 26
- Batch Size: 16
- Validation Split: 30%

Data Augmentation:

- Horizontal Flip
- Zoom
- Shear
- Rescaling

Model saved as: ***xcep_yoga.h5***

System Architecture

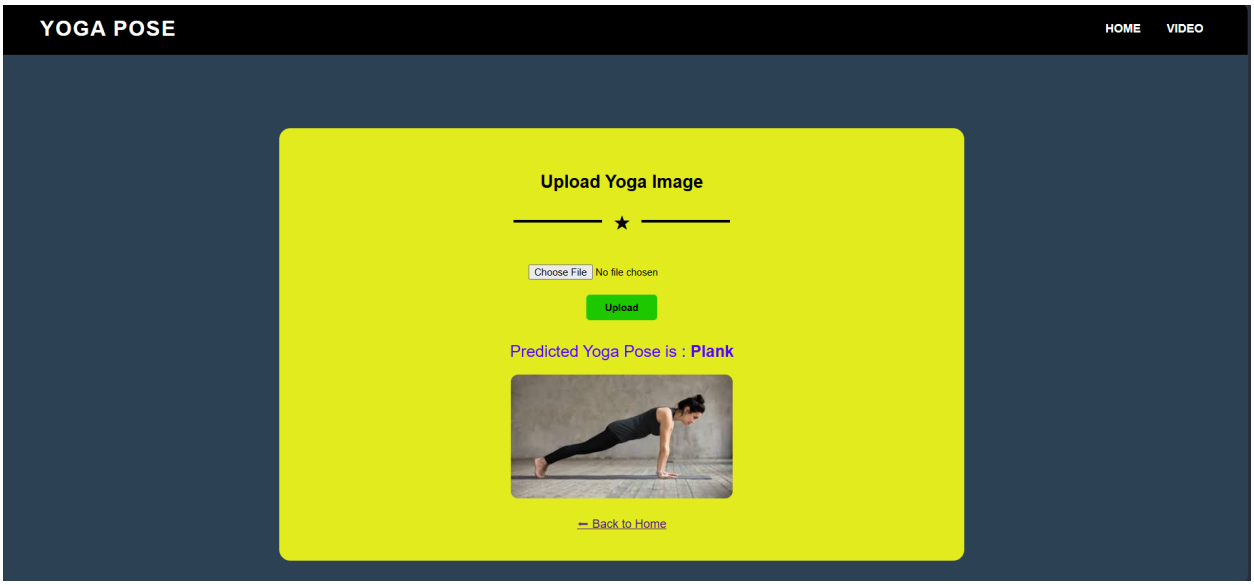


Web Application Demo

The system is deployed using a **Flask web application**.

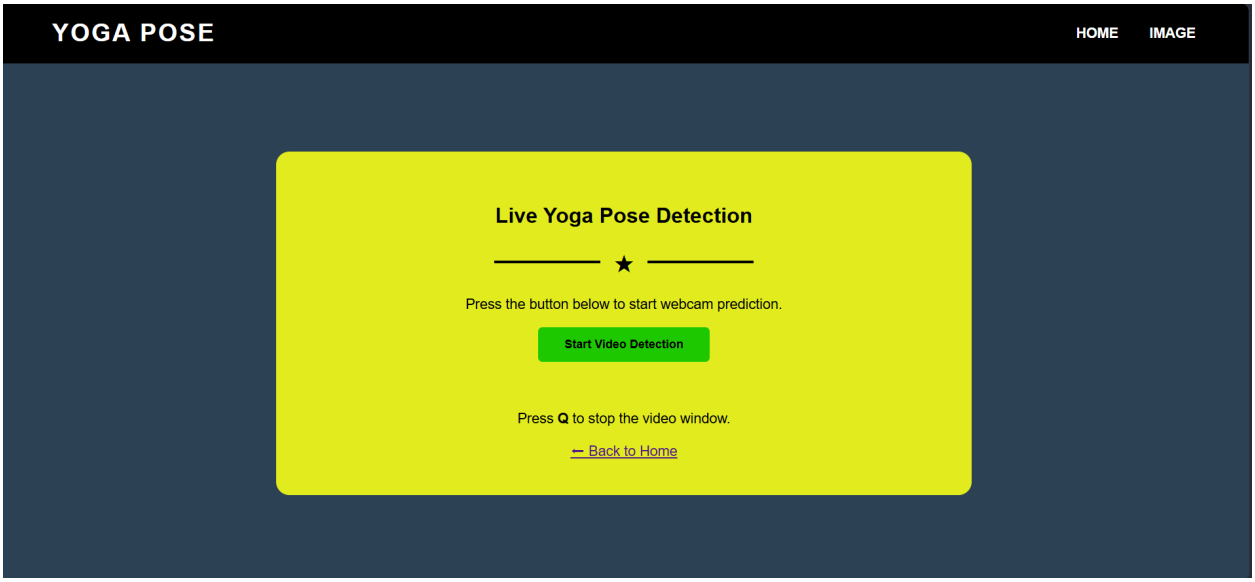
Image Prediction

- Upload yoga pose image
- Model predicts pose class



Video Detection

- Start webcam
- Real-time pose prediction



Watch a live demo of ZenFlow in action:

➤ [Click here to watch the video]

Conclusion & Future Work

Conclusion:

- Built a yoga pose classification system
- Used **Transfer Learning (Xception)**
- Achieved accurate predictions
- Developed a **Flask web application**

Future Improvements:

- Add more yoga poses
- Improve dataset size
- Deploy on cloud
- Add posture correction feedback

Thank You!